🔎 Topic 1: Computer Networks Basics + Network Lab

Abhi aap kahan hain?

Aap apne computer par baithay hain. Aapko ek virtual lab banana hai jahan aap Windows aur Linux dono experiment kar sakain.

Step 1: VirtualBox Install Karein

- Google par jayen: "VirtualBox download"

- Apne OS ke mutabiq installer download karein aur install karein.

Step 2: Pehli Virtual Machine (Windows Server) Banayein

- VirtualBox open karein > **New** button press karein.

- Name: `Windows-Server`

- Folder: C:/Users/YourName/VirtualBox VMs/ (default)

- ISO Image: Browse karein aur Windows Server ISO file select karein.

- Skip Unattended Installation (checkbox unchecked rakhein).

- Next press karein.

- Memory (RAM): `4096 MB` (4GB) dein.

- Next > Processors: `2` dein.

- Next > Hard Disk: `50 GB` dein.

- Next > Finish.

Step 3: Doosri Virtual Machine (Linux) Banayein

- New button press karein.

- Name: `Linux-Server`

- ISO Image: Ubuntu Server ya CentOS ISO select karein.

- Memory: `2048 MB` (2GB)

- Hard Disk: `20 GB`

- Finish.

Step 4: Network Setting (Dono VMs ko ek network mein laana)

- Windows-Server VM par right-click > Settings > Network.
- Adapter 1: **NAT** (internet ke liye)
- Adapter 2: **Host-Only** (dono VMs aapas mein baat kar sakain) Enable karein.
- Linux-Server VM par bhi yahi setting karein.

Lab Task: Dono VMs install karein. Install hone ke baad dono ko ek saath chalayein.

🔎 Topic 2: Server Environment — Windows & Linux

Abhi aap kahan hain?

Windows Server aur Linux Server dono install ho chukay hain. Ab aap dono ko compare kar rahe hain.

Step 1: Windows Server Interface Dekhein

- Windows Server VM on karein.
- Login karein (Administrator / password jo aap ne set kiya tha).
- **Server Manager** khulega. Yeh Windows ka central control panel hai.
- Left side mein: Dashboard, Local Server, All Servers, etc.
- **Local Server** click karein. Yahan aapko computer ka naam, domain, IP address dikhega.

Step 2: Linux Server Interface Dekhein

- Linux Server VM on karein.
- Login karein (username/password).
- Terminal khulega (black screen). Yeh Linux ka command-line interface hai.
- `whoami` type karein > Enter. Output: `admin` (current user batayega).
- `pwd` type karein > Enter. Output: `/home/admin` (current folder batayega).

Step 3: Network Setting Check Karein (Windows)

- Windows mein Command Prompt open karein (Start > cmd).

- `ipconfig` type karein.

- Output mein dekhein: `Ethernet adapter Ethernet 2` ke neechay `IP Address: 192.168.xx.xx` (Host-Only network ki IP).

Step 4: Network Setting Check Karein (Linux)

- Linux terminal mein: `ifconfig` type karein.

- `eth0` = NAT (internet), `eth1` = Host-Only.

- `eth1` ki IP note karein (jaise `192.168.56.101`).

Lab Task: Dono VMs ki Host-Only IPs note karein. Windows se Linux ko ping karein:

- Windows CMD: `ping 192.168.56.101` (Linux ki IP)

- Agar reply aata hai toh dono VMs connected hain.

🔑 Topic 3: Client/Server OS Installation & Configuration

Abhi aap kahan hain?

Aap ne Windows aur Linux install kar liye hain. Ab aap unhein "configure" kar rahe hain taake woh server ki tarah behave kar sakain.

Configuration ka matlab: Server ko ek static (fixed) IP dena taake uska address kabhi change na ho.

Step 1: Windows Server Ko Static IP Dena

- Windows Server VM mein: Server Manager > Local Server.

- `Ethernet` (jis par Host-Only hai) par click karein.

- Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties.

- "Use the following IP address" select karein.

- IP address: `192.168.56.10`

- Subnet mask: `255.255.255.0`

- Default gateway: leave blank (Host-Only ke liye).

- DNS server: `127.0.0.1` (khud ko) daalein.

- OK > Close.

Step 2: Linux Server Ko Static IP Dena (Ubuntu)

- Linux terminal mein: `sudo nano /etc/netplan/00-installer-config.yaml`

- Ye file edit karein:

```yaml
network:
  ethernets:
    eth1:
      dhcp4: false
      addresses: [192.168.56.11/24]
  version: 2
```

- Ctrl+O > Enter (save) > Ctrl+X (exit).

- `sudo netplan apply` (settings apply karein).

- `ifconfig` se check karein ke IP set ho gayi.

Lab Task: Windows ki IP `192.168.56.10`, Linux ki IP `192.168.56.11` set karein. Phir Windows se Linux ko ping karein: `ping 192.168.56.11` - ping aani chahiye.

🔎 Topic 4: Linux Installation & Verification

Abhi aap kahan hain?

Linux install ho chuka hai aur IP set ho chuki hai. Ab aap system ko "verify" kar rahe hain ke sab theek hai.

Step 1: Operating System Version Check

- Linux terminal mein: `uname -a`

- Output: `Linux Linux-Server 5.4.0-26-generic #30-Ubuntu SMP Mon Apr 20 16:58:30 UTC 2020 x86_64 x86_64 x86_64 GNU/Linux`

- **Samajh:** Kernel version `5.4.0-26`, architecture `x86_64` (64-bit).

Step 2: Hard Disk Space Check

- `df -h`

- Output:

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       20G   5G   15G  25% /
```

- **Samajh:** Total disk 20GB hai, 5GB used, 15GB free.

Step 3: Memory (RAM) Check

- `free -m`

- Output:

```
              total   used   free   shared   buff/cache   available
Mem:          1995    400   1200      10          395        1400
```

- **Samajh:** Total RAM 1995 MB (~2GB), used 400MB, free 1200MB.

Step 4: Network Connectivity Check

- `ping -c 4 8.8.8.8`

- Agar output aata hai `64 bytes from 8.8.8.8` toh internet chal raha hai.

Step 5: Disk Partitions Check

- `lsblk`

- Output:

```
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda      8:0    0   20G  0 disk
├─sda1   8:1    0   19G  0 part /
├─sda2   8:2    0    1G  0 part [SWAP]
```

- **Samajh:** Main partition `/` (root) 19GB hai, SWAP 1GB hai.

Lab Task: Yeh sab commands likhein aur output copy karein. Verify karein ke hard disk, RAM, network sab green (OK) hain.

🔑 Topic 5: Active Directory + Domain Controller

Abhi aap kahan hain?

Windows Server par kaam kar rahe hain. Ab isko Domain Controller (DC) banana hai. DC poori network ka "Boss" hota hai.

Step 1: Active Directory Role Install Karein

- Windows Server VM mein: Server Manager open karein.

- **Add Roles and Features** click karein.

- Next > Next > (Server Selection) Next.

- Server Roles mein: **Active Directory Domain Services** tick karein.

- "Add Features" pop-up aayega > Add Features click karein.

- Next > Next > Next > Install.

- Installation complete hone tak wait karein.

Step 2: Domain Controller Promote Karein

- Installation ke baad, Server Manager mein ek flag icon (▲) aayega.

- Us par click karein > **Promote this server to a domain controller**.

- "Add a new forest" select karein.

- Root domain name: `mycompany.local` likhein.

- Next > Domain Controller Options:

  - Password: `Admin@123` (set karein)

  - Confirm password: `Admin@123`

- Next > Next > (DNS options) Next.

- Additional Options: NetBIOS domain name: `MYCOMPANY` (auto aayega).

- Next > Paths (default rakhein) > Next.

- Review > Install.

- System reboot ho jayega.

Step 3: Verify Domain Controller

- Reboot ke baad login karein.

- Server Manager > Tools > **Active Directory Users and Computers** open karein.

- Left side mein `mycompany.local` dikhega. Yeh aapka domain hai.

Lab Task: Active Directory Users and Computers open karein. Domain tree expand karein. `Users` folder, `Computers` folder dekhein.

🔎 Topic 6: Active Directory Forest & Domain Management

Abhi aap kahan hain?

Domain Controller ban chuka hai. Ab aap forest aur domain ki structure bana rahe hain.

Step 1: Organizational Units (OUs) Banana

- Active Directory Users and Computers mein:

- `mycompany.local` par right-click > New > Organizational Unit.

- Name: `Sales` likhein > OK.

- Dobara right-click > New > Organizational Unit.

- Name: `IT` likhein > OK.

- Name: `HR` likhein > OK.

Step 2: OUs Ke Andar Sub-OUs Banana

- `Sales` par right-click > New > Organizational Unit.

- Name: `Karachi` likhein > OK.

- `Sales` par right-click > New > Organizational Unit.

- Name: `Lahore` likhein > OK.

Step 3: Structure Dekhein

- Ab tree kuch aisa dikhega:

```
mycompany.local (Domain)
├── Builtin (folder)
├── Computers (folder)
├── Domain Controllers (folder)
├── Sales (OU)
|   ├── Karachi (OU)
```

| └── Lahore (OU)

├── IT (OU)

└── HR (OU)

```

**Step 4: Computers Ko OUs Mein Move Karein**

- `Computers` folder mein jo computers hain (jaise `WIN-ABC123`), unhein:

- `WIN-ABC123` par right-click > Move.

- `Sales` OU select karein > OK.

**Lab Task:** OUs banayein aur Computers folder mein se computer ko Sales OU mein move karein.

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## 🔑 Topic 7: Users & Groups / Domain Controller Administration

**Abhi aap kahan hain?**

Aap Active Directory mein users aur groups bana rahe hain.

**Step 1: User Banana (Ali)**

- Active Directory Users and Computers mein:

- `Sales` OU par right-click > New > User.

- First name: `Ali`

- User logon name: `ali`

- Next > Password: `P@ssw0rd` > Confirm: `P@ssw0rd`

- "User must change password" **uncheck** karein.

- "Password never expires" **check** karein.

- Next > Finish.
```

Step 2: User Banana (Ahmed)

- `IT` OU par right-click > New > User.

- First name: `Ahmed`

- User logon name: `ahmed`

- Password: `P@ssw0rd` > Finish.

Step 3: Group Banana (Sales-Team)

- `Sales` OU par right-click > New > Group.

- Group name: `Sales-Team`

- Group scope: `Global` (select)

- Group type: `Security` (select)

- OK.

Step 4: User Ko Group Mein Add Karna

- `Ali` par right-click > Add to a group.

- Object Names: `Sales-Team` likhein > Check Names > OK.

- OK (confirmation aayega).

Step 5: Group Members Check Karein

- `Sales-Team` par double-click > Members tab.

- `Ali` dikhega.

PowerShell Commands (Option):

- PowerShell (Admin) open karein.

- `New-ADUser -Name "Zain" -UserPrincipalName "zain@mycompany.local" -GivenName "Zain" -Surname "Khan" -Enabled $true -AccountPassword (ConvertTo-SecureString "P@ssw0rd" -AsPlainText -Force)`

- `New-ADGroup -Name "IT-Team" -GroupScope Global`

- `Add-ADGroupMember -Identity "IT-Team" -Members "Zain"`

Lab Task: Users aur groups banayein. Ali ko Sales-Team mein add karein.

🔎 Topic 8: Network Monitoring Tools

Abhi aap kahan hain?

Aap Linux server par hain. Network monitor kar rahe hain.

Step 1: Connectivity Check (ping)

- `ping -c 4 192.168.56.10` (Windows server ki IP)

- Output:

```
64 bytes from 192.168.56.10: icmp_seq=1 ttl=128 time=0.5 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=128 time=0.4 ms
```

- **Samajh:** Windows server alive hai aur response de raha hai.

Step 2: Network Interfaces Check (ifconfig)

- `ifconfig`

- Output:

```
eth0: flags=4163<UP,BROADCAST,RUNNING>  mtu 1500
      inet 10.0.2.15  netmask 255.255.255.0
eth1: flags=4163<UP,BROADCAST,RUNNING>  mtu 1500
      inet 192.168.56.11  netmask 255.255.255.0
```

- **Samajh:** `eth0` = NAT (internet), `eth1` = Host-Only.

Step 3: Ports/Listening Services Check (ss)

- `ss -tuln`

- Output:

```
Netid  State   Recv-Q  Send-Q  Local Address:Port   Peer Address:Port
tcp    LISTEN  0      128     0.0.0.0:22          0.0.0.0:*
tcp    LISTEN  0      128     0.0.0.0:80          0.0.0.0:*
```

- **Samajh:** Port 22 (SSH) open hai, Port 80 (HTTP) open hai.

Step 4: Active Connections Check (netstat)

- `netstat -tulpn`

- Output:

```
Proto Recv-Q Send-Q Local Address   Foreign Address  State   PID/Program
tcp   0     0     0.0.0.0:22      0.0.0.0:*       LISTEN  1234/sshd
tcp   0     0     0.0.0.0:80      0.0.0.0:*       LISTEN  5678/apache2
```

Step 5: Route Table Check

- `route -n`

- Output:

```
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         10.0.2.2        0.0.0.0         UG    100    0        0 eth0
192.168.56.0    0.0.0.0         255.255.255.0   U     0      0        0 eth1
```

- **Samajh:** Default gateway `10.0.2.2` (internet ke liye).

Lab Task: Yeh sab commands run karein aur output dekhein. Port 22 aur 80 open hain ya nahi check karein.

🔑 Topic 9: Boot Management

Abhi aap kahan hain?

Linux server par hain. Boot process ko control kar rahe hain.

Step 1: Boot Time Check

- `systemd-analyze`

- Output: `Startup finished in 2.345s (kernel) + 8.567s (userspace) = 10.912s`

- **Samajh:** System boot hone mein ~11 seconds lage.

Step 2: Sab Se Slow Services Check

- `systemd-analyze blame`

- Output:

```
5.123s network.service

3.456s apache2.service

1.234s ssh.service
```

- **Samajh:** `network.service` sab se zyada time le raha hai boot mein.

Step 3: Service Enable/Disable (Boot par auto-start)

- `sudo systemctl enable apache2` (Apache ko boot par start hone do)

- Output: `Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service`

- `sudo systemctl disable apache2` (Apache ko boot par start hone se roko)

- Output: `Removed /etc/systemd/system/multi-user.target.wants/apache2.service`

Step 4: Sari Services Dekhein

- `systemctl list-units --type=service`

- Output: Sari active services ki list.

Step 5: System Reboot

- `sudo reboot` (system restart ho jayega)

- Reboot ke baad check karein: `systemctl status apache2` (agar enable kiya tha toh running hogi)

Lab Task: Apache service ko enable karein, reboot karein, aur check karein ke boot par start ho gayi.

🔑 Topic 10: Process Management

Abhi aap kahan hain?

Linux server par hain. Running processes ko monitor aur control kar rahe hain.

Step 1: Sari Processes Dekhein (Snapshot)

- `ps aux`

- Output (sample):

```
USER       PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 168948 10248 ?        Ss   10:00   0:02 /sbin/init
admin     1234  0.5  1.2 234567 24567 ?        S    10:05   0:05 /usr/bin/python3
```

- **Samajh:** `PID` = Process ID, `%CPU` = CPU usage, `%MEM` = Memory usage.

Step 2: Live Process Monitor (top)

- `top`
- Live update hota rahega. Press `q` to exit.
- Columns: `PID`, `USER`, `PR`, `NI`, `VIRT`, `RES`, `SHR`, `S`, `%CPU`, `%MEM`, `TIME+`, `COMMAND`

Step 3: Process Ko Forcefully Close Karna

- Pehle `top` mein kisi process ka PID note karein (jaise 1234).
- `kill 1234` (gracefully close)
- Agar process hang ho: `kill -9 1234` (forcefully close)

Step 4: Process Ki Detail Dekhein

- `ps -p 1234 -o pid,ppid,cmd`
- Output: PID, parent PID, aur command dikhega.

Step 5: Background Process Chalana

- `sleep 100 &` (sleep command ko background mein chalao)
- `jobs` (background mein chal rahi jobs dekho)
- `fg %1` (job ko foreground mein lao)

Lab Task: `top` karo, kisi process ka PID note karo, aur `kill` karo.

🎯 Topic 11: IP Tables + Packet Filtering

Abhi aap kahan hain?

Linux server par firewall configure kar rahe hain.

Step 1: Current Rules Dekhein

- `sudo iptables -L`

- Output:

```
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
```

- **Samajh:** Abhi koi rule nahi hai, sab traffic allow hai.

Step 2: SSH (Port 22) Allow Karein

- `sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT`

- **Samajh:** INPUT chain mein rule add karo, TCP protocol, destination port 22, accept karo.

Step 3: HTTP (Port 80) Allow Karein

- `sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT`

Step 4: HTTPS (Port 443) Allow Karein

- `sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT`

Step 5: Baqi Sab Block Karein

- `sudo iptables -A INPUT -j DROP`

- **Samajh:** INPUT chain mein aakhri rule: baqi sab drop (block) karo.

Step 6: Rules Verify Karein

- `sudo iptables -L -v -n`

- Output:

```
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 ACCEPT     tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:22
    0     0 ACCEPT     tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:80
    0     0 ACCEPT     tcp  --  *      *       0.0.0.0/0            0.0.0.0/0           tcp dpt:443
    0     0 DROP       all  --  *      *       0.0.0.0/0            0.0.0.0/0
```

- **Samajh:** Port 22,80,443 allow hain, baqi sab drop.

Step 7: Rules Delete Karein (Flush)

- `sudo iptables -F` (sari rules delete)

- `sudo iptables -L` (check karein, ab kuch nahi)

Lab Task: SSH, HTTP allow karein, baqi sab block karein. Phir Windows se Linux ko ping karein (ping block ho jani chahiye).

🎤 Topic 12: Securing Network Traffic

Abhi aap kahan hain?

Aap network traffic ko encrypt kar rahe hain taake koi beech mein na dekh sake.

Step 1: SSL Certificate Generate Karein

- `openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout server.key -out server.crt`
- Questions poochhega:
  - Country Name: `PK`
  - State: `Sindh`
  - City: `Karachi`
  - Organization: `MyCompany`
  - Common Name: `192.168.56.11` (apni IP daalein)
- Output: `server.key` aur `server.crt` files ban jayengi.

Step 2: SSH Service Secure Karein
- `sudo nano /etc/ssh/sshd_config`
- Ye lines change karein:
  - `PermitRootLogin no` (root ko login se roko)
  - `PasswordAuthentication yes` (password se login allow)
  - `Port 22` (default port)
- `sudo systemctl restart sshd`

Step 3: SSH Se Connect Karein (Windows se)
- Windows CMD: `ssh admin@192.168.56.11`
- Password daalein, login ho jayen ge.

Step 4: Firewall Mein SSH Allow Karein (Agar block hai toh)
- `sudo ufw allow 22` (UFW firewall hai toh)
- `sudo ufw enable`

Step 5: Secure Copy (SCP)
- Windows se Linux par file copy:
  - `scp C:\Users\YourName\file.txt admin@192.168.56.11:/home/admin/`
- Linux se Windows par file copy:

- `scp admin@192.168.56.11:/home/admin/file.txt C:\Users\YourName\`

Lab Task: SSL certificate generate karein. SSH enable karein aur Windows se Linux par SSH karein.

🔑 Topic 13: Advanced File Systems & Logs

Abhi aap kahan hain?

Aap file systems aur logs dekh rahe hain.

Step 1: File System Type Check

- `df -T`

- Output:

```
Filesystem     Type     1K-blocks    Used Available Use% Mounted on
/dev/sda1      ext4      20480000 5000000  14480000  26% /
```

- **Samajh:** File system `ext4` hai.

Step 2: Disks aur Partitions Dekhein

- `lsblk -f`

- Output:

```
NAME   FSTYPE LABEL UUID                                 MOUNTPOINT
sda
├─sda1 ext4         1234-5678-90ab-cdef                  /
├─sda2 swap         8765-4321-fedc-ba98                  [SWAP]
```

Step 3: System Logs Dekhein (journalctl)

- `journalctl` (sare logs)

- `journalctl -n 20` (last 20 lines)

- `journalctl -u sshd` (sirf SSH ke logs)

- Output:

```
Jun 15 10:05:34 Linux-Server sshd[1234]: Accepted password for admin from 192.168.56.10 port 54321 ssh2
```

- **Samajh:** Admin ne SSH se login kiya.

Step 4: Real-time Logs Dekhein

- `tail -f /var/log/syslog` (Ubuntu)

- `tail -f /var/log/messages` (CentOS)

- Live logs aate rahenge. Press `Ctrl+C` to exit.

Step 5: Log Rotation (Logs clean karna)

- `sudo logrotate -f /etc/logrotate.conf` (logs rotate karein)

Lab Task: `journalctl -u sshd` karke dekhein ke kaun logins hain. `tail -f /var/log/syslog` karke live logs dekhein.

🎤 Topic 14: Bash Shell Scripting

Abhi aap kahan hain?

Aap ek script likh rahe hain jo kaam automatically karegi.

Step 1: Pehli Script Banayein (Hello World)

- `nano hello.sh`

- Yeh likhein:

```bash
#!/bin/bash
echo "Hello World! Yeh meri pehli script hai."
date
whoami
```

- Ctrl+O > Enter (save) > Ctrl+X (exit).

Step 2: Script Ko Executable Banayein

- `chmod +x hello.sh`

Step 3: Script Run Karein

- `./hello.sh`

- Output:

```
Hello World! Yeh meri pehli script hai.
Mon Jun 15 10:30:00 PKT 2026
admin
```

Step 4: Backup Script Banayein

- `nano backup.sh`

- Yeh likhein:

```bash
#!/bin/bash
```

```
# Backup Script
SOURCE="/home/admin/Documents"
DEST="/home/admin/Backup"
DATE=$(date +%Y-%m-%d)
echo "Backup start: $DATE"
mkdir -p $DEST
cp -r $SOURCE/* $DEST/
echo "Backup complete! Files copied to $DEST"
ls -la $DEST
```

- `chmod +x backup.sh`
- `./backup.sh`

Step 5: Variables Aur User Input

- `nano vars.sh`

```bash
#!/bin/bash
echo "Apna naam likhein:"
read NAME
echo "Assalam-o-Alaikum, $NAME! Aaj ki tareekh hai:"
date
```

- `./vars.sh` run karein.

Step 6: Loop (For) Script

- `nano loop.sh`

```bash
#!/bin/bash
for i in 1 2 3 4 5
```

```
do
  echo "Number: $i"
done

for file in /home/admin/*
do
  echo "File: $file"
done
```

- `./loop.sh` run karein.

Step 7: If-Else Script

- `nano if.sh`

```bash
#!/bin/bash
echo "Enter a number:"
read NUM
if [ $NUM -gt 10 ]
then
  echo "$NUM is greater than 10"
else
  echo "$NUM is less than or equal to 10"
fi
```

- `./if.sh` run karein.

Lab Task: Backup script likhein, run karein, aur verify karein ke files copy ho gayi hain.

🔎 Topic 15.1: FTP Server

Abhi aap kahan hain?

Aap Linux par FTP server install kar rahe hain taake files transfer kar saken.

Step 1: FTP Server Install Karein

- `sudo apt update` (Ubuntu)
- `sudo apt install vsftpd -y`
- `sudo systemctl start vsftpd`
- `sudo systemctl enable vsftpd`

Step 2: FTP Server Status Check

- `sudo systemctl status vsftpd`
- Output: `active (running)` dikhna chahiye.

Step 3: FTP Configuration

- `sudo nano /etc/vsftpd.conf`
- Ye lines change karein / uncomment karein:
  - `anonymous_enable=NO` (anonymous users ko roko)
  - `local_enable=YES` (local users ko allow karo)
  - `write_enable=YES` (upload ki permission do)
  - `local_umask=022` (file permissions set karo)
- `sudo systemctl restart vsftpd`

Step 4: FTP User Banana

- `sudo useradd -m ftpuser`
- `sudo passwd ftpuser` (password set karein: `ftp123`)

Step 5: Windows Se FTP Connect Karein

- Windows CMD: `ftp 192.168.56.11`

- Username: `ftpuser` > Password: `ftp123`

- `dir` (files dekhein)

- `get filename.txt` (file download karein)

- `put filename.txt` (file upload karein)

- `bye` (exit)

Step 6: FileZilla Use Karein

- FileZilla install karein (Windows par).

- Host: `192.168.56.11`, Username: `ftpuser`, Password: `ftp123`, Port: `21`

- Connect karein aur files drag-drop karein.

Lab Task: FTP install karein, ftpuser banayein, Windows se FTP connect karein aur file upload/download karein.

🎤 Topic 15.2: NFS Server

Abhi aap kahan hain?

Aap NFS (Linux-to-Linux file sharing) configure kar rahe hain.

Step 1: NFS Server Install Karein

- `sudo apt install nfs-kernel-server -y`

Step 2: Share Folder Banayein

- `sudo mkdir /srv/nfs_share`

- `sudo chown nobody:nogroup /srv/nfs_share`

- `sudo chmod 777 /srv/nfs_share`

Step 3: Exports File Edit Karein

- `sudo nano /etc/exports`
- Ye line add karein:
  - `/srv/nfs_share 192.168.56.0/24(rw,sync,no_subtree_check)`
- **Samajh:** `/srv/nfs_share` ko 192.168.56.x network ke sabko read-write allow karo.

Step 4: NFS Restart Karein

- `sudo exportfs -a` (exports apply karein)
- `sudo systemctl restart nfs-kernel-server`

Step 5: Doosre Linux VM Par Client Side

- Doosri Linux VM (client) par: `sudo apt install nfs-common -y`
- `sudo mkdir /mnt/nfs_client`
- `sudo mount 192.168.56.11:/srv/nfs_share /mnt/nfs_client`
- `df -h` (mount check karein)
- `cd /mnt/nfs_client` > `touch test.txt` (file create karein)
- Server par jaake dekhein: `/srv/nfs_share` mein file aagayi.

Lab Task: NFS share banayein, doosri VM se mount karein, file create karein aur verify karein.

🎤 Topic 15.3: Samba Server

Abhi aap kahan hain?

Aap Samba (Windows-Linux file sharing) configure kar rahe hain.

Step 1: Samba Install Karein

- `sudo apt install samba smbclient cifs-utils -y`

Step 2: Share Folder Banayein

- `sudo mkdir /srv/samba_share`
- `sudo chmod 777 /srv/samba_share`

Step 3: Samba Config File Edit Karein

- `sudo nano /etc/samba/smb.conf`
- End par yeh add karein:

```
[Share]
  path = /srv/samba_share
  browseable = yes
  read only = no
  guest ok = no
  valid users = smbuser
```

Step 4: Samba User Banana

- `sudo useradd smbuser`
- `sudo smbpasswd -a smbuser` (password set karein: `smb123`)
- `sudo smbpasswd -e smbuser` (user enable karein)

Step 5: Samba Restart Karein

- `sudo systemctl restart smbd`
- `sudo systemctl enable smbd`

Step 6: Windows Se Samba Connect Karein

- Windows File Explorer open karein.
- Address bar mein: `\\192.168.56.11\Share`
- Username: `smbuser`, Password: `smb123`
- Files copy/paste karein.

Step 7: Linux Se Samba Connect Karein (Client)
- `sudo smbclient //192.168.56.11/Share -U smbuser`
- `ls` (files dekhein)
- `get file.txt` (download)
- `put file.txt` (upload)

Lab Task: Samba share banayein, Windows se connect karein aur files share karein.

🔑 Topic 15.4: DHCP Server

Abhi aap kahan hain?
Aap DHCP server configure kar rahe hain jo IP addresses automatically dega.

Step 1: DHCP Server Install Karein
- `sudo apt install isc-dhcp-server -y`

Step 2: DHCP Config File Edit Karein
- `sudo nano /etc/dhcp/dhcpd.conf`
- End par yeh add karein:
```
subnet 192.168.56.0 netmask 255.255.255.0 {
  range 192.168.56.100 192.168.56.200;
```

```
  option routers 192.168.56.1;
  option domain-name-servers 8.8.8.8, 8.8.4.4;
  default-lease-time 600;
  max-lease-time 7200;
}
```

Step 3: Interface Set Karein
- `sudo nano /etc/default/isc-dhcp-server`
- `INTERFACESv4="eth1"` (Host-Only interface)

Step 4: DHCP Start Karein
- `sudo systemctl start isc-dhcp-server`
- `sudo systemctl enable isc-dhcp-server`
- `sudo systemctl status isc-dhcp-server`

Step 5: Client Side (Windows)
- Windows VM mein network adapter ki setting "Obtain IP automatically" karein.
- DHCP se IP mil jayegi (192.168.56.100 - 200 range mein).

Step 6: DHCP Leases Check Karein
- `sudo cat /var/lib/dhcp/dhcpd.leases`
- Output mein client ki IP dikhegi.

Lab Task: DHCP configure karein, Windows client ko DHCP se IP mile.

🎤 Topic 15.5: DNS Server

Abhi aap kahan hain?

Aap DNS server configure kar rahe hain jo domain names ko IP mein convert karega.

Step 1: DNS Server (Bind9) Install Karein

- `sudo apt install bind9 -y`

Step 2: DNS Config File Edit Karein

- `sudo nano /etc/bind/named.conf.local`
- End par yeh add karein:

```
zone "mywebsite.com" {
   type master;
   file "/etc/bind/db.mywebsite.com";
};
```

Step 3: Zone File Banayein

- `sudo cp /etc/bind/db.local /etc/bind/db.mywebsite.com`
- `sudo nano /etc/bind/db.mywebsite.com`
- Edit karein:

```
$TTL    604800
@       IN      SOA     mywebsite.com. root.mywebsite.com. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
```

;

@ IN NS ns.mywebsite.com.

@ IN A 192.168.56.11

ns IN A 192.168.56.11

www IN A 192.168.56.11

```

- **Samajh:** `www.mywebsite.com` ko `192.168.56.11` se map kiya.

**Step 4: DNS Start Karein**

- `sudo systemctl start bind9`

- `sudo systemctl enable bind9`

**Step 5: Client Side (Windows)**

- Windows VM mein DNS server set karein: `192.168.56.11`

- CMD: `nslookup www.mywebsite.com`

- Output: `Address: 192.168.56.11`

**Step 6: DNS Logs Check Karein**

- `sudo tail -f /var/log/syslog | grep named`

**Lab Task:** DNS configure karein, client par `nslookup` karein aur domain name resolve ho.

---

## 🔎 Topic 15.6: Apache/Web Server

**Abhi aap kahan hain?**

Aap web server (Apache) install kar rahe hain taake website host kar saken.
```

Step 1: Apache Install Karein

- `sudo apt install apache2 -y`

Step 2: Apache Start Karein

- `sudo systemctl start apache2`
- `sudo systemctl enable apache2`
- `sudo systemctl status apache2`

Step 3: Website Create Karein

- `sudo nano /var/www/html/index.html`
- Yeh likhein:

```html
<!DOCTYPE html>
<html>
<head><title>Meri Pehli Website</title></head>
<body>
<h1>Assalam-o-Alaikum!</h1>
<p>Yeh meri pehli website hai. Apache server chal raha hai!</p>
</body>
</html>
```

Step 4: Firewall Mein Port 80 Allow Karein

- `sudo ufw allow 80` (UFW hai toh)
- `sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT` (iptables hai toh)

Step 5: Windows Se Website Access Karein

- Windows browser mein: `http://192.168.56.11`
- Website dikhegi.

Step 6: Virtual Host Banayein (Doosri Website)

- `sudo mkdir /var/www/site2`
- `sudo nano /var/www/site2/index.html`
- `<h1>Site 2</h1>` likhein.
- `sudo nano /etc/apache2/sites-available/site2.conf`

```
<VirtualHost *:80>
    ServerName site2.local
    DocumentRoot /var/www/site2
</VirtualHost>
```

- `sudo a2ensite site2.conf`
- `sudo systemctl reload apache2`

Lab Task: Apache install karein, website create karein, browser se access karein.

🔎 Topic 16: Managing Basic System Issues + Troubleshooting

Abhi aap kahan hain?

Aap system issues diagnose kar rahe hain.

Step 1: Disk Full Issue Check

- `df -h`
- Agar `/` 100% full hai toh: `sudo du -sh /* | sort -h` (kaunsi folder sab se zyada space le rahi hai)

Step 2: Memory Issue Check

- `free -m`

- Agar swap use ho raha hai toh memory kam hai: `top` press `M` (memory ke hisaab se sort)

Step 3: Service Down Issue Check

- `sudo systemctl status apache2`

- Agar inactive hai toh: `sudo systemctl start apache2`

- Error logs: `sudo journalctl -u apache2 -n 20`

Step 4: Network Issue Check

- `ping 192.168.56.10` (Windows server)

- `ping 8.8.8.8` (internet)

- `ip route` (routing table check)

Step 5: Port Already In Use Check

- `sudo ss -tulpn | grep 80`

- Agar port 80 use ho raha hai: `sudo lsof -i :80` (kaunsi process use kar rahi hai)

Step 6: Permission Issue Check

- `ls -la /var/www/html/index.html`

- Agar permission nahi hai: `sudo chmod 644 /var/www/html/index.html`

Step 7: Logs Check

- `sudo journalctl -xe` (sare errors)

- `sudo tail -f /var/log/syslog` (live logs)

Lab Task: Apache stop karein (`sudo systemctl stop apache2`), troubleshoot karein, restart karein.

🔎 Topic 17: Wireless Access Point & Topic 18: Wireless Router

Abhi aap kahan hain?

Cisco Packet Tracer open karein. (Software download karein agar nahi hai)

Step 1: Wireless Router Drag Karein

- Packet Tracer open karein.
- Left side > Network Devices > Wireless Devices.
- `Linksys WRT300N` drag karein main screen par.

Step 2: PC Add Karein

- End Devices > PC drag karein (2 PCs).
- Dono PCs mein Wireless NIC daalein:
  - PC > Physical tab > Left side `WMP300N` drag karein empty slot mein.
  - Power off/on karein.

Step 3: Wireless Router Configure Karein

- Wireless Router par click > GUI tab.
- Setup > Internet Setup:
  - IP Address: `192.168.0.1`
  - Subnet Mask: `255.255.255.0`
- Wireless > Basic Wireless Settings:
  - SSID: `MyWiFi`
  - Channel: `6`
- Wireless > Wireless Security:
  - Security Mode: `WPA2-Personal`
  - Pre-Shared Key: `Password123`

Step 4: PCs Ko WiFi Se Connect Karein

- PC > Desktop > PC Wireless.

- `MyWiFi` select karein > Connect.

- Password: `Password123`

- `ipconfig` se check karein IP mil gayi (192.168.0.x)

Step 5: Ping Check

- PC1 se PC2 ko ping karein: `ping 192.168.0.3` (PC2 ki IP)

Lab Task: Wireless router configure karein, PCs connect karein, ping check karein.

🔎 Topic 19: Cisco Switches — Manageable/Non-manageable

Abhi aap kahan hain?

Packet Tracer mein manageable switch configure kar rahe hain.

Step 1: Switch Drag Karein

- Network Devices > Switches.

- `2960` switch drag karein (yeh manageable hai).

- `2950T` bhi manageable hai.

- `Hub` non-manageable hai (kuch configure nahi kar sakte).

Step 2: Switch Configure Karein (CLI)

- Switch par click > CLI tab.

- `enable` (privileged mode)

- `configure terminal` (global config)

- `hostname SW1` (name set karein)

- `enable secret cisco` (enable password set karein)

Step 3: VLAN Create Karein

- `vlan 10`
- `name Sales`
- `vlan 20`
- `name IT`

Step 4: Ports Configure Karein

- `interface fastEthernet 0/1`
- `switchport mode access`
- `switchport access vlan 10`

Step 5: Show Commands

- `show vlan brief`
- `show running-config`
- `show interfaces status`

Step 6: Save Configuration

- `copy running-config startup-config`
- `write memory`

Lab Task: Switch configure karein, VLANs banayein, ports assign karein.

🎯 Topic 20: VLAN + Switch Modes

Abhi aap kahan hain?

Aap VLANs configure kar rahe hain.

Step 1: VLAN Configuration (Same as above)

- `enable`
- `configure terminal`
- `vlan 10`
- `name Sales`
- `vlan 20`
- `name IT`

Step 2: Access Mode Ports Configure

- `interface fastEthernet 0/1`
- `switchport mode access`
- `switchport access vlan 10`
- `interface fastEthernet 0/2`
- `switchport mode access`
- `switchport access vlan 20`

Step 3: Trunk Mode (Multiple VLANs carry karne ke liye)

- `interface fastEthernet 0/24`
- `switchport mode trunk`
- `switchport trunk allowed vlan 10,20`

Step 4: Testing

- PC1 (VLAN 10) ko PC2 (VLAN 10) ping karein -> ping aani chahiye.
- PC1 (VLAN 10) ko PC3 (VLAN 20) ping karein -> ping nahi aani chahiye.

Step 5: Show VLANs

- `show vlan brief`

Lab Task: VLANs banayein, PCs assign karein, ping test karein.

🎤 Topic 21: WAN + WAN Simulator

Abhi aap kahan hain?

Aap WAN simulate kar rahe hain (do cities ko jodna).

Step 1: Two Routers Add Karein

- Network Devices > Routers.
- `2911` drag karein (Router 1 - Lahore)
- Doosra `2911` drag karein (Router 2 - Karachi)

Step 2: Serial Connection (WAN Link)

- Connections > Serial DCE cable select karein.
- Router 1 ka `Serial 0/0/0` se Router 2 ka `Serial 0/0/0` connect karein.

Step 3: Router 1 (Lahore) Configure

- `enable`
- `configure terminal`
- `interface Serial 0/0/0`
- `ip address 10.0.0.1 255.255.255.252`
- `clock rate 64000` (DCE side par)
- `no shutdown`
- `interface GigabitEthernet 0/0`
- `ip address 192.168.1.1 255.255.255.0`
- `no shutdown`

Step 4: Router 2 (Karachi) Configure

- `enable`
- `configure terminal`
- `interface Serial 0/0/0`
- `ip address 10.0.0.2 255.255.255.252`
- `no shutdown`
- `interface GigabitEthernet 0/0`
- `ip address 192.168.2.1 255.255.255.0`
- `no shutdown`

Step 5: PCs Add Karein

- Router 1 ke sath PC1 (192.168.1.10/24, gateway 192.168.1.1)
- Router 2 ke sath PC2 (192.168.2.10/24, gateway 192.168.2.1)

Lab Task: Routers configure karein, WAN link set karein, PCs connect karein.

🎯 Topic 22: Static Routing

Abhi aap kahan hain?

Aap static routes daal rahe hain taake routers ek doosre ke networks ko jaan sakein.

Step 1: Router 1 (Lahore) Static Route

- `enable`
- `configure terminal`
- `ip route 192.168.2.0 255.255.255.0 10.0.0.2`
- **Samajh:** 192.168.2.0 network ke liye 10.0.0.2 (Router 2) par jao.

Step 2: Router 2 (Karachi) Static Route

- `enable`
- `configure terminal`
- `ip route 192.168.1.0 255.255.255.0 10.0.0.1`

Step 3: Verify Routes

- `show ip route`
- Output mein `S` (static) routes dikhenge.

Step 4: Ping Test

- PC1 se PC2 ping karein: `ping 192.168.2.10`
- Ping successful honi chahiye.

Lab Task: Static routes daalein, ping test karein.

🎤 Topic 23: Dynamic Routing — RIP

Abhi aap kahan hain?

Aap RIP dynamic routing configure kar rahe hain (routers apne aap routes share karein).

Step 1: Router 1 (Lahore) Par RIP Configure

- `enable`
- `configure terminal`
- `router rip`
- `version 2` (RIPv2 use karein)
- `network 192.168.1.0`
- `network 10.0.0.0`

- `no auto-summary`

Step 2: Router 2 (Karachi) Par RIP Configure

- `enable`
- `configure terminal`
- `router rip`
- `version 2`
- `network 192.168.2.0`
- `network 10.0.0.0`
- `no auto-summary`

Step 3: Verify RIP Routes

- `show ip route`
- Output mein `R` (RIP) routes dikhenge.

Step 4: Debug RIP (Live Updates Dekhein)

- `debug ip rip`
- Routes exchange hoti dekhein.
- `no debug ip rip` (debug off)

Step 5: Ping Test

- PC1 se PC2 ping karein.

Lab Task: RIP configure karein, routes exchange dekhein, ping test karein.

🔑 Topic 24: IGRP (Interior Gateway Routing Protocol)

Abhi aap kahan hain?

Aap IGRP (Cisco proprietary) dynamic routing configure kar rahe hain.

Step 1: Router 1 (Lahore) Par IGRP Configure

- `enable`
- `configure terminal`
- `router igrp 100` (100 = Autonomous System number)
- `network 192.168.1.0`
- `network 10.0.0.0`

Step 2: Router 2 (Karachi) Par IGRP Configure

- `enable`
- `configure terminal`
- `router igrp 100`
- `network 192.168.2.0`
- `network 10.0.0.0`

Step 3: Verify IGRP Routes

- `show ip route`
- Output mein `I` (IGRP) routes dikhenge.

Step 4: Show Protocols

- `show ip protocols`
- IGRP details dekhein (metric, updates).

Lab Task: IGRP configure karein, routes verify karein.

🔎 Topic 25: ACL Configuration on Routers

Abhi aap kahan hain?

Aap Access Control List (ACL) bana rahe hain taake specific traffic ko block/allow kar saken.

Step 1: Standard ACL Banayein (Source IP basis)

- Router 1 (Lahore) par:

- `enable`

- `configure terminal`

- `access-list 10 deny 192.168.1.0 0.0.0.255` (VLAN 10 wale ko block karo)

- `access-list 10 permit any` (baqi sab allow)

Step 2: ACL Ko Interface Par Apply Karein

- `interface GigabitEthernet 0/0`

- `ip access-group 10 out` (outbound traffic par apply)

Step 3: Extended ACL Banayein (Source + Destination + Port)

- `access-list 100 deny tcp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 eq 80` (HTTP ko block)

- `access-list 100 permit ip any any`

Step 4: Interface Par Apply Karein

- `interface Serial 0/0/0`

- `ip access-group 100 in`

Step 5: Show ACLs

- `show access-lists`

- `show access-lists 10`

Step 6: ACL Remove Karein

- `no access-list 10`

- `no ip access-group 10 out` (interface se hatao)

Lab Task: ACL banayein, apply karein, test karein ke block ho gaya.

🎯 Topic 26: Complete Practical/Viva Revision

Ab aap kahan hain?

Final revision kar rahe hain. Apne saare lab tasks review karein.

Step 1: Sab Commands Likhein (Cheat Sheet)

Linux Commands:

```
uname -a    = System info
df -h       = Disk space
free -m     = RAM check
ping        = Connectivity
ifconfig    = IP address
ss -tuln    = Open ports
systemctl   = Service management
iptables    = Firewall
```

Cisco Commands:

```
enable                  = Privileged mode
```

configure terminal = Config mode

show ip route = Routing table

show vlan brief = VLANs

ip route = Static route

router rip = RIP configure

access-list = ACL banayein

```

**Windows Commands (PowerShell):**

```

ipconfig = IP check

ping = Connectivity

New-ADUser = User banayein

Add-ADGroupMember = Group mein add karein

```

**Step 2: Viva Questions Prepare Karein**

**Q1: VLAN kyun banate hain?**

A: Network ko chhote logical groups mein todne ke liye, security aur performance improve karne ke liye.

**Q2: Static aur Dynamic routing mein farq?**

A: Static manually karte hain, dynamic routers apne aap routes seekhte hain (RIP, OSPF).

**Q3: DHCP ka kya kaam hai?**

A: Automatically IP addresses distribute karna.

**Q4: DNS kya karta hai?**

A: Domain names ko IP addresses mein resolve karta hai.
```

Q5: ACL kyun use karte hain?

A: Network traffic ko filter karne ke liye (allow/deny).

Q6: Apache kya hai?

A: Web server jo websites host karta hai.

Q7: Samba aur NFS mein farq?

A: Samba Windows-Linux share, NFS Linux-Linux share.

Q8: IP Tables kya hai?

A: Linux ka firewall.

Q9: Active Directory kya hai?

A: Microsoft ki directory service jo users, computers manage karti hai.

Q10: Boot management kya hai?

A: System start-up process ko control karna.

Step 3: Practical Demo Ready Rakhein

- Linux VM open rakhein.
- `systemctl status apache2` dikhayein.
- `iptables -L` dikhayein.
- `ping` connectivity dikhayein.
- Apache website browser mein dikhayein.